

HS 26 Environmental Studies

ENVIRONMENT, ECOLOGY AND BIODIVERSITY



ENVIRONMENT, ECOLOGY AND BIODIVERSITY

Environmental science

Environmental= Living + Non-living things.

- Study of the environment (Biotic and Abiotic components)

TYPES OF ENVIRONMENT

- Natural environment

- ❖ Soil, water, air, trees, noise, etc.,

- Man-made environment

- ❖ House, road, park, etc.,

Components of environment

- Abiotic or non-living components
 - Biotic or living components
 - Energy components

ABIOTIC COMPONENTS

- Atmosphere
- Lithosphere
- Hydrosphere

Functions of atmosphere

- It sustains life on the earth.
- It observes the cosmic rays and other electromagnetic radiation coming from the sun.
- It balances heat of earth by absorption of IR and UV rays.
- It plays important role in carrying water from the ocean to the land through hydrological cycle.
- Oxygen supports living beings, CO_2 is essential for photosynthesis of plants.

Structure of atmosphere

- Troposphere (0 – 18kms)
- Stratosphere (18 – 50kms)
- Mesosphere (50 – 85kms)
- Thermosphere (or) ionosphere (85 – 500kms)
- Exosphere (upto 1600kms)

Energy components

- Energy flow across biotic and abiotic components.

E.g:

- solar energy,
- nuclear energy,
- geochemical energy,
- thermo electrical energy, etc.,

Scope of environmental studies

- To get awareness and sensitivity to the total environment.
- To motivate the active participation in environmental protection and improvement.
- To know the necessity of conservation of natural resources .
- To evaluate environmental programmes in terms of social, economic, ecological and aesthetic factors.

Need for public awareness

- Man has overexploited the natural resource which leads to many environmental problems such as acid rain, ozone layer depletion, green house effect, landslides, cancer, and other health problems.
- Lack of awareness and less number of people participation leads to poor pollution management which leads to climate instability and unhealthy eco-system

A photograph of a calm lake in a forest. The water reflects the surrounding green trees and a fallen log. A large, gnarled tree trunk leans over the water on the right. The word "ECO-SYSTEM" is written in large, yellow, serif capital letters across the middle of the image.

ECO-SYSTEM

Stiesta Lake, Yosemite National Park, California

ECOSYSTEM

- An ecosystem is formed by the interaction between all living and non-living things.

BIOME

- Set of ecosystem which are exposed to same climatic conditions with similar life cycle, climatic conditions and physical structure.

❖ **Biome (small ecosystem).**

ABIOTIC COMPONENTS

- Solar energy provides practically all the energy for ecosystems.
- Inorganic substances, e.g., sulfur, boron, tend to cycle through ecosystems.
- Organic compounds, such as proteins, carbohydrates, lipids, and other complex molecules, form a link between biotic and abiotic components of the system.

BIOTIC COMPONENTS

living organisms of an ecosystem.

Classification of Biotic components are:

AUTOTROPHS

Organisms that produce their own food from an energy source, such as the sun, and inorganic compounds.

HETEROTROPHS

Organisms that consume other organisms as a food source.

MEMBERS OF BIOTIC COMPONENTS

❖ PRODUCERS.

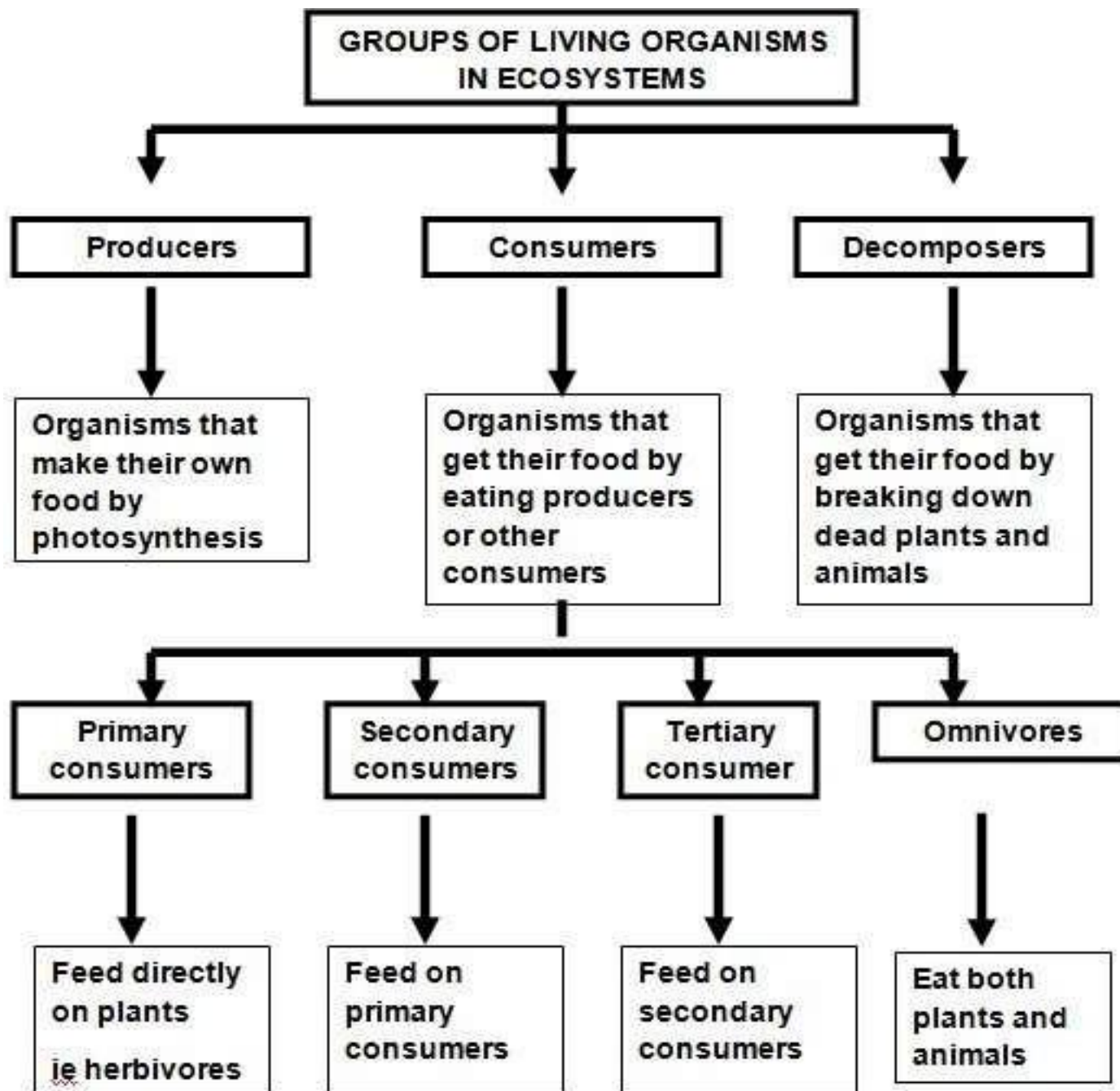
❖ CONSUMERS.

➤ Primary Consumers

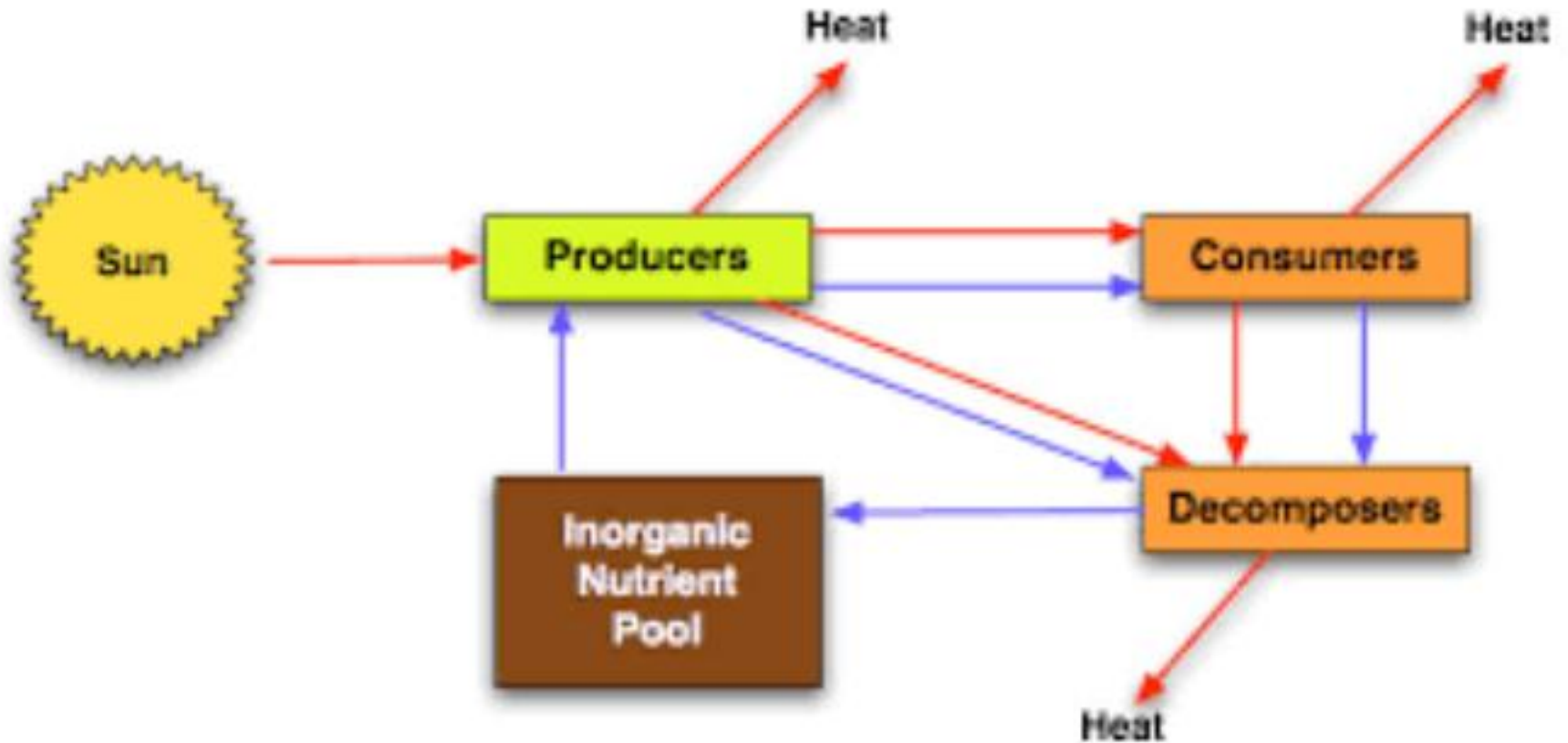
➤ Secondary Consumers

➤ Tertiary Consumers

❖ DECOMPOSERS.



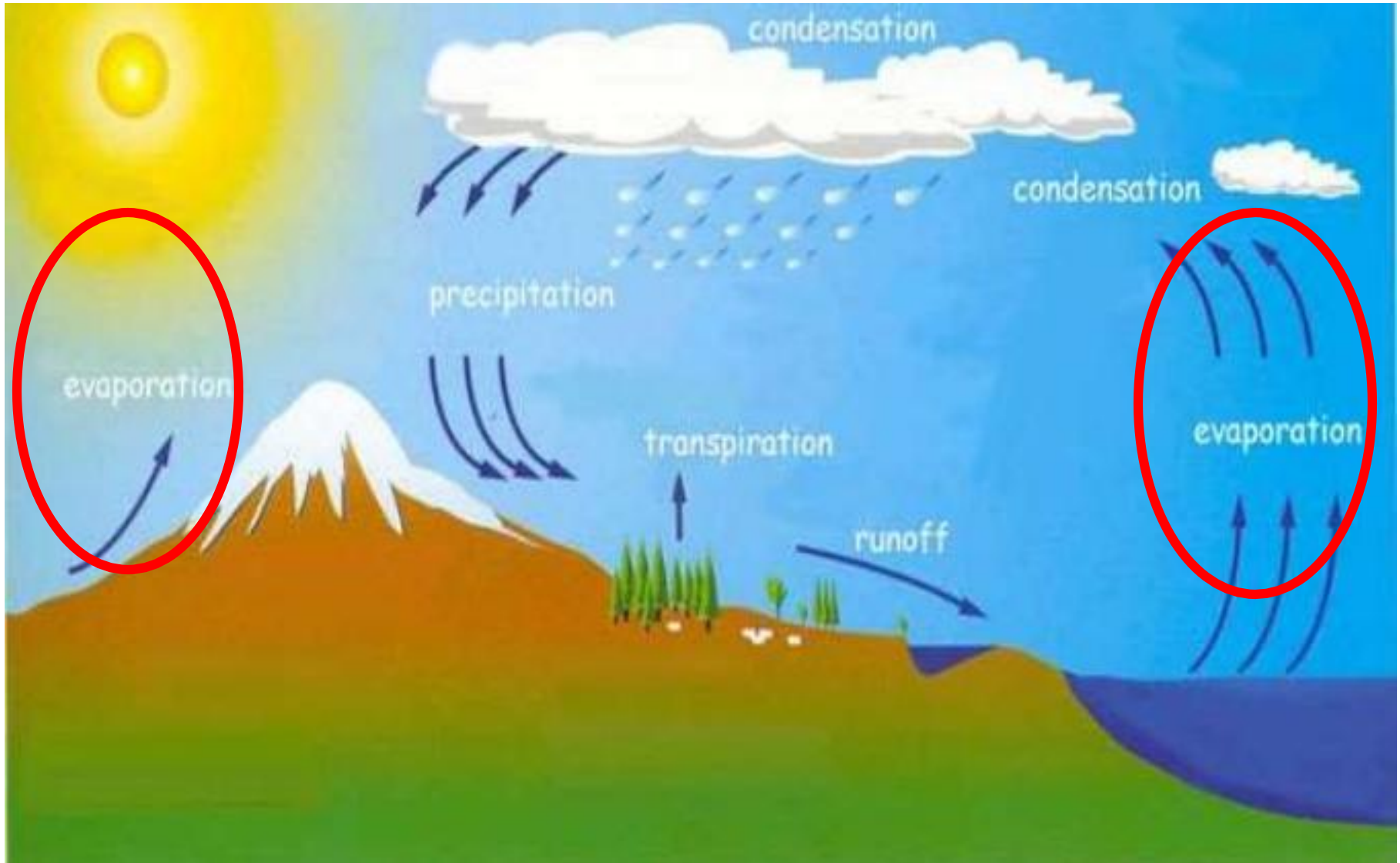
ENERGY AND MATERIAL FLOW IN ECOSYSTEM



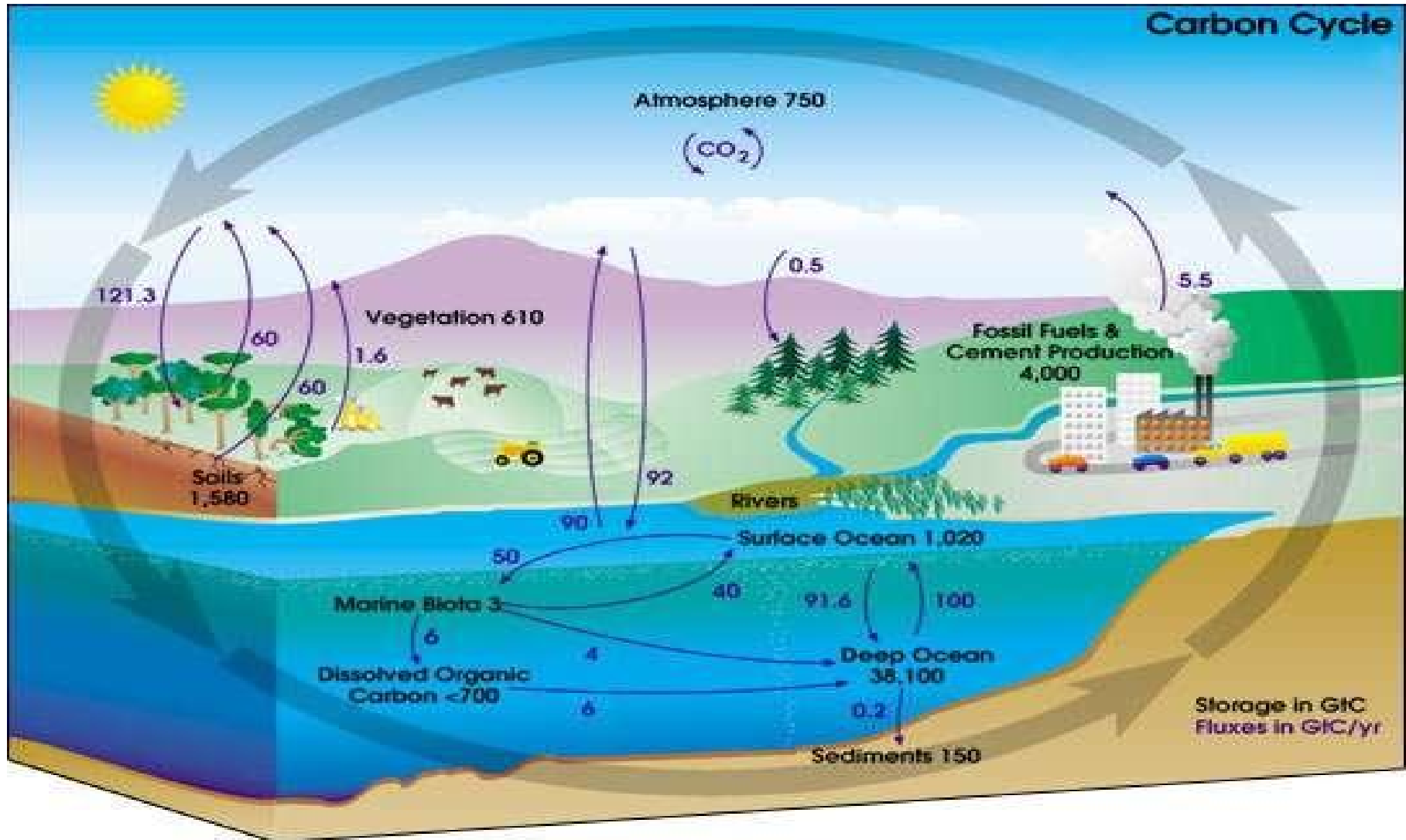
ENERGY FLOW IN ECOSYSTEM

- Energy is defined as the capacity to do work. For living organisms, it is the basic force responsible for running all the metabolic activities. The flow of energy from producer level to top consumer level is called energy flow.
- The flow of energy in an ecosystem is unidirectional. It flows from producer level to consumer level.
- The process of energy flow involves transfer of energy from autotrophs to various components of heterotrophs and help in maintaining bio diversity.
- The main source of energy in the ecosystem is sunlight. About 80% of energy is lost during flow of energy from one trophic level to the next one.

The Water Cycle

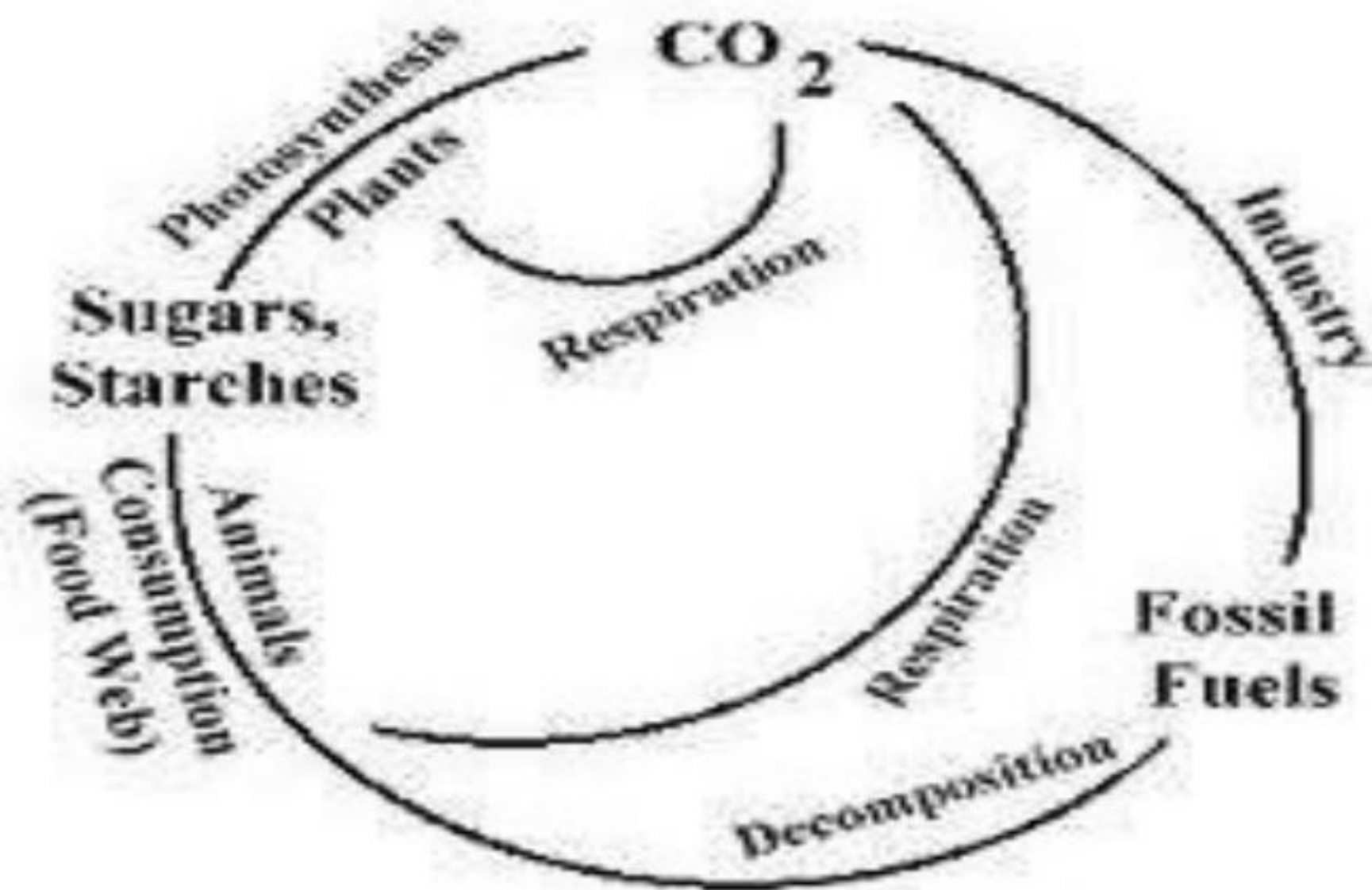


Global Carbon Cycle

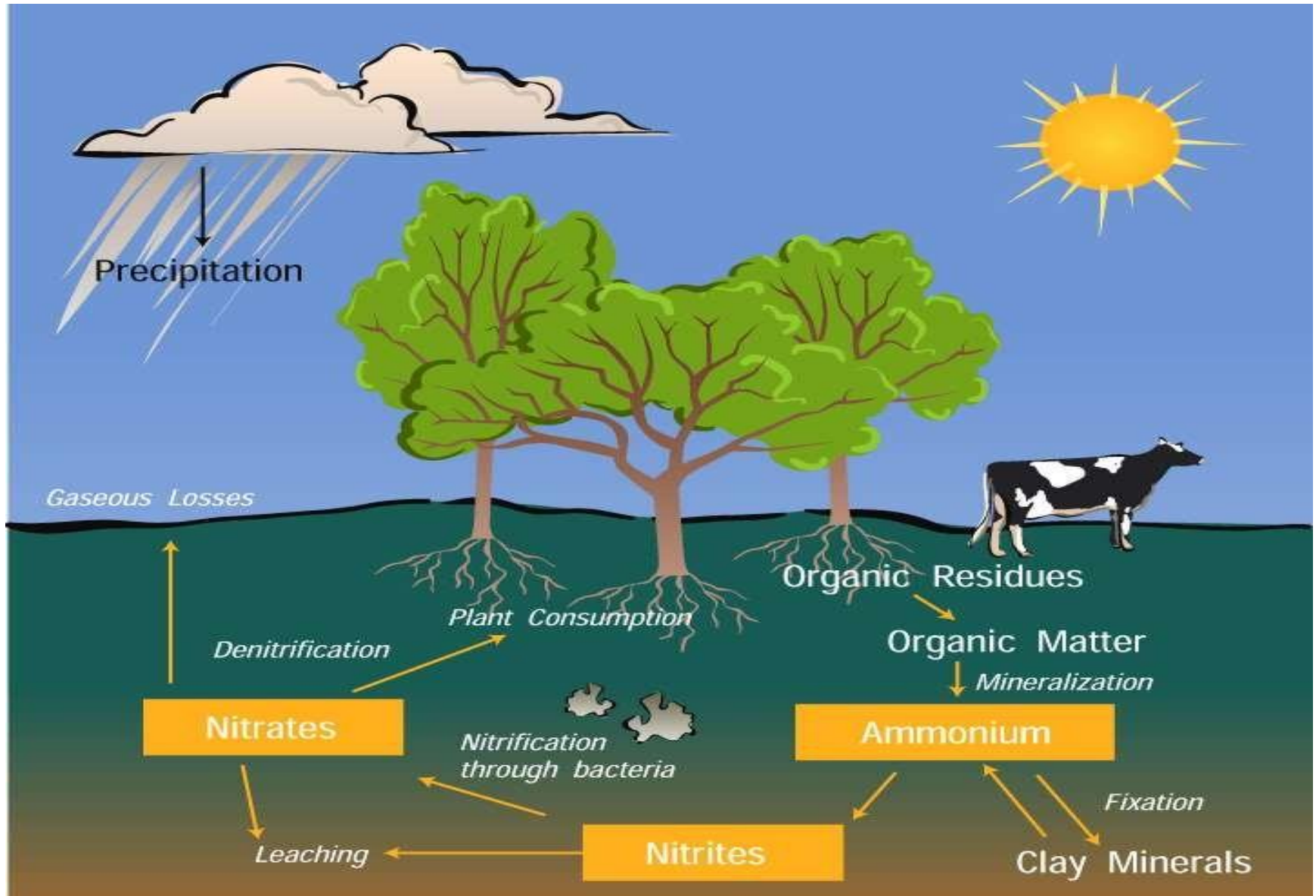


Carbon is exchanged between the active pools due to various processes – photosynthesis and respiration between the land and the atmosphere, and diffusion between the ocean and the atmosphere.

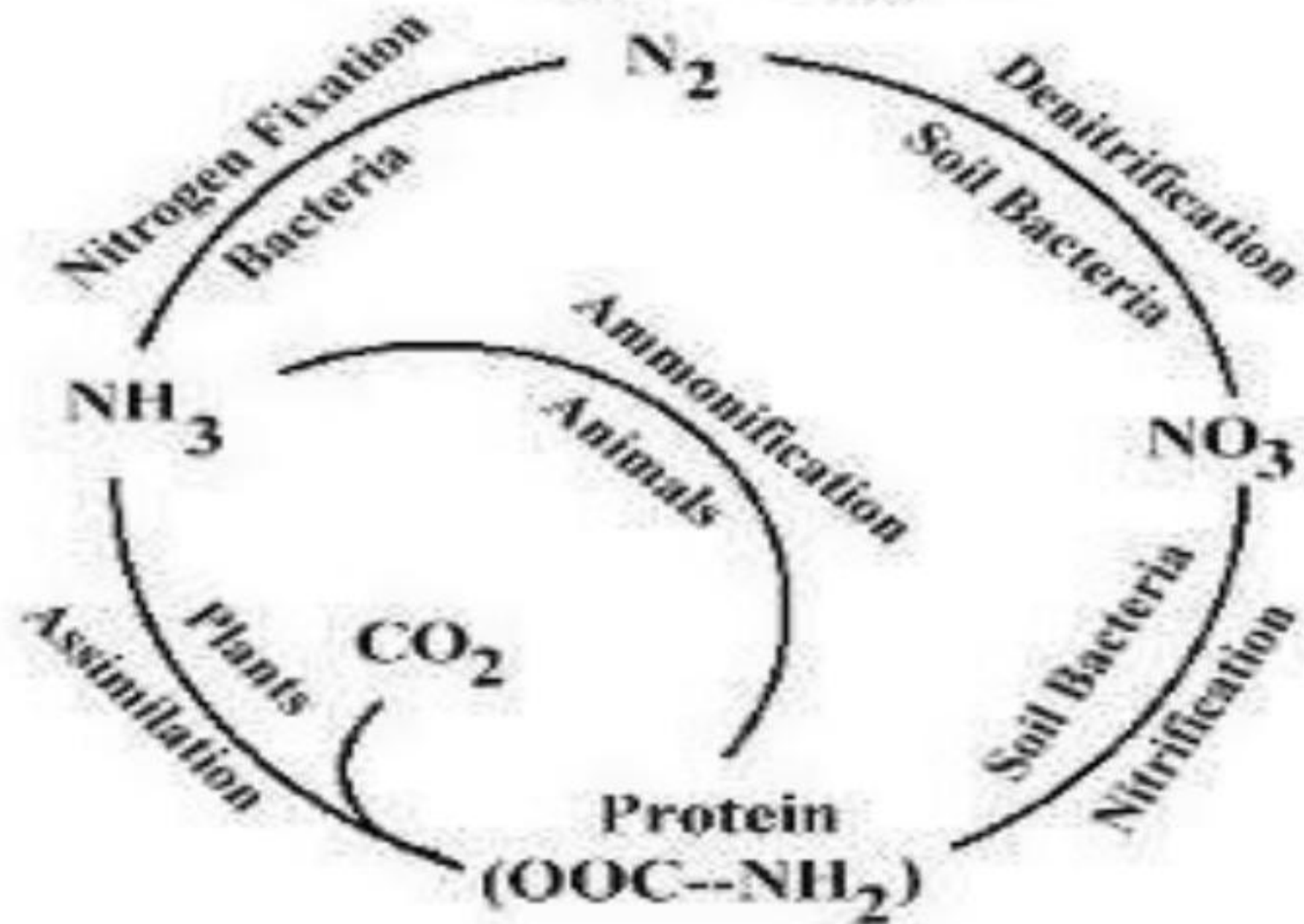
Carbon Cycle

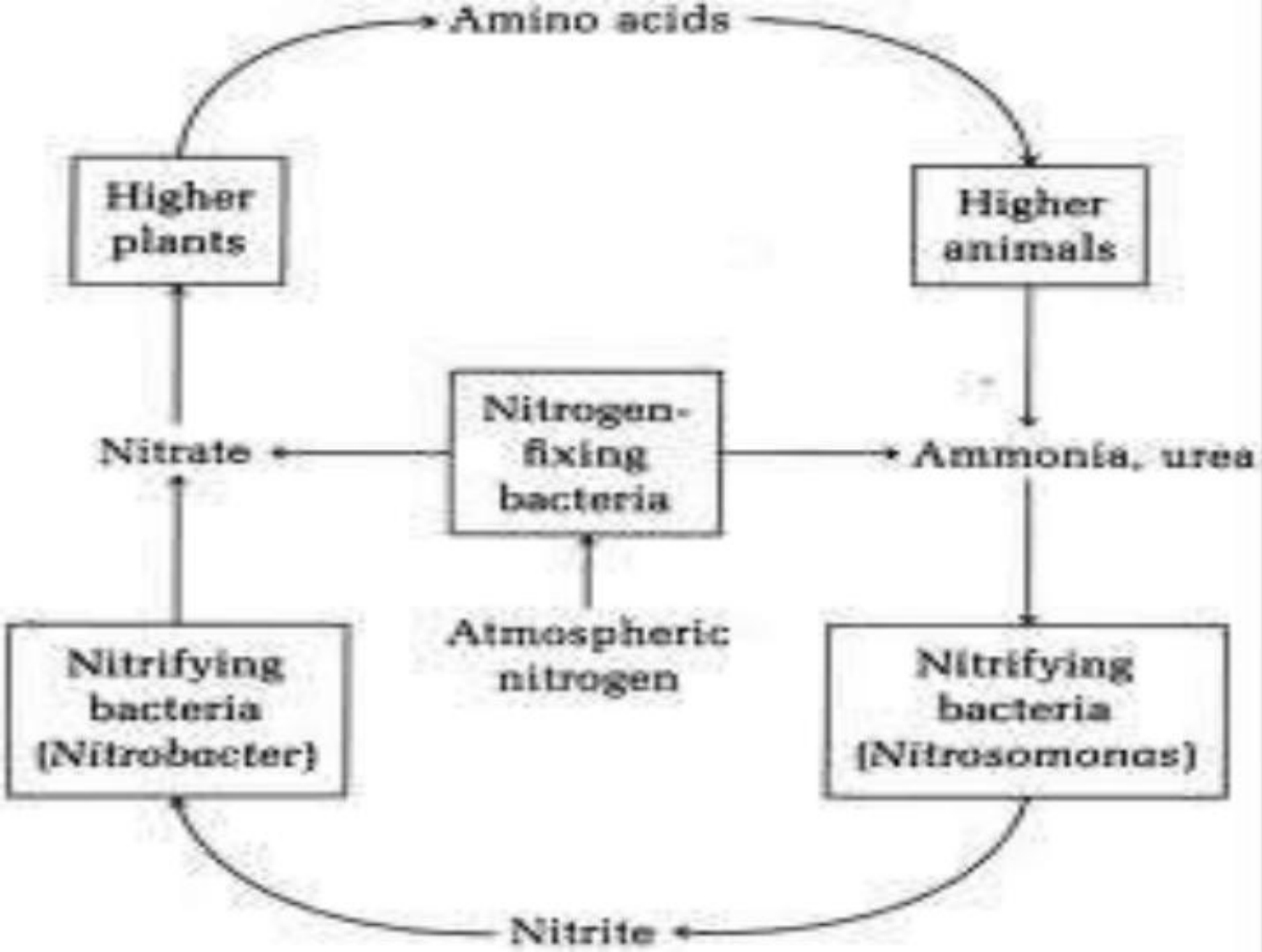


The N cycle over land



Nitrogen Cycle





The P cycle

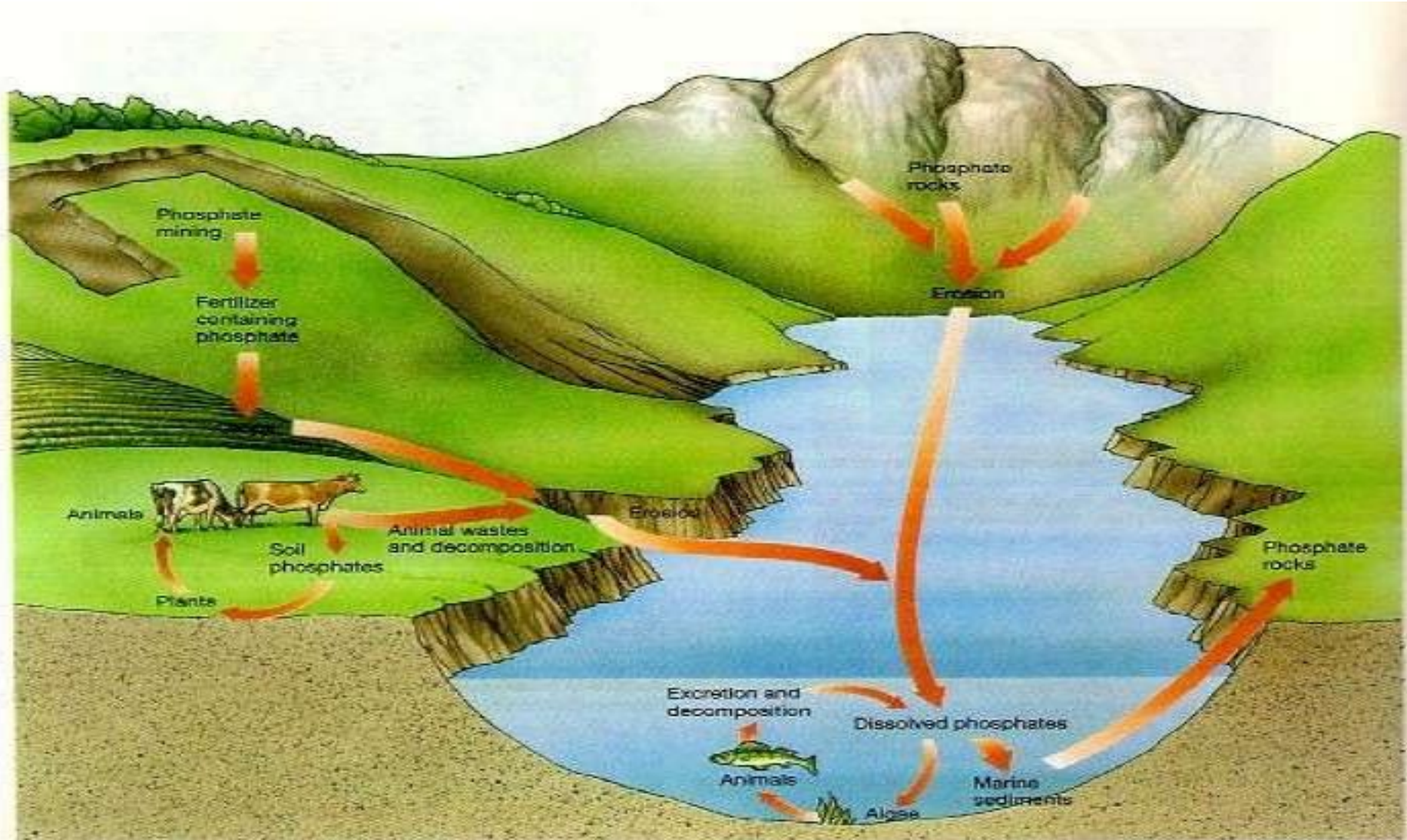
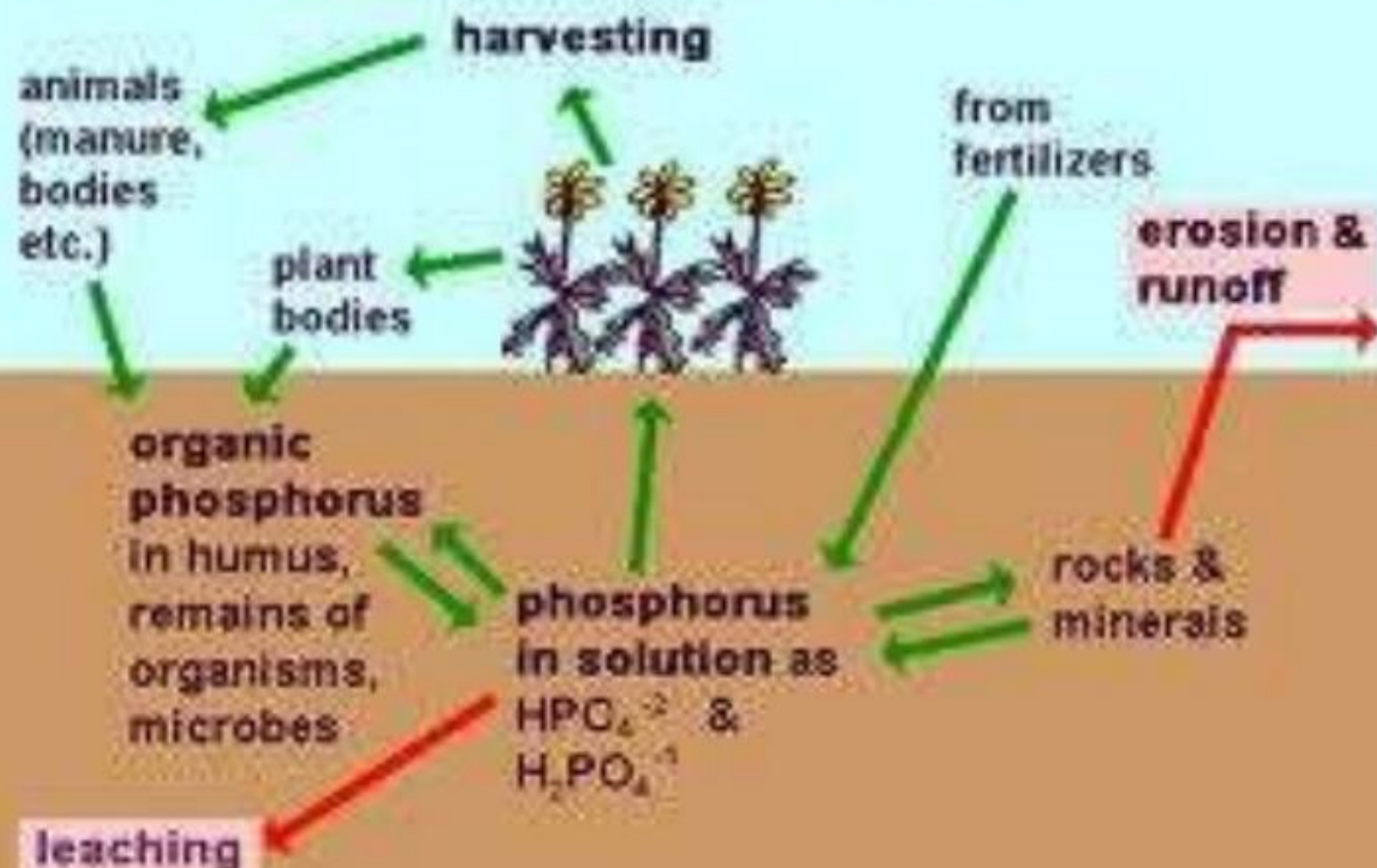


Figure 54-5 The phosphorus cycle in terrestrial and aquatic environments. Recycling of phosphorus (as phosphate, PO_4^{3-}) is slow because no biologically important form of phosphorus is

gaseous. Phosphate that becomes part of marine sediments may take millions of years to solidify into rock, uplift as mountains, and erode again to become available to living things.

The Phosphorus Cycle



Phosphorus Cycle

Rocks &
Soil

Run-Off
Lakes & Streams

Inorganic
Phosphate

Consumption
Animals

Plants
Absorption

PO_4^{3-}

ATP

Decomposition

Metabolism

Tropic Levels

A tropic level is the position occupied by an organism in a food chain. Tropic levels can be analyzed on an energy pyramid.

- **Producers - First tropic level.**
- **Primary consumers - Second tropic level.**
- **Secondary consumers - Third tropic level.**
- **Tertiary consumers - Top tropic level.**

Trophic
Levels

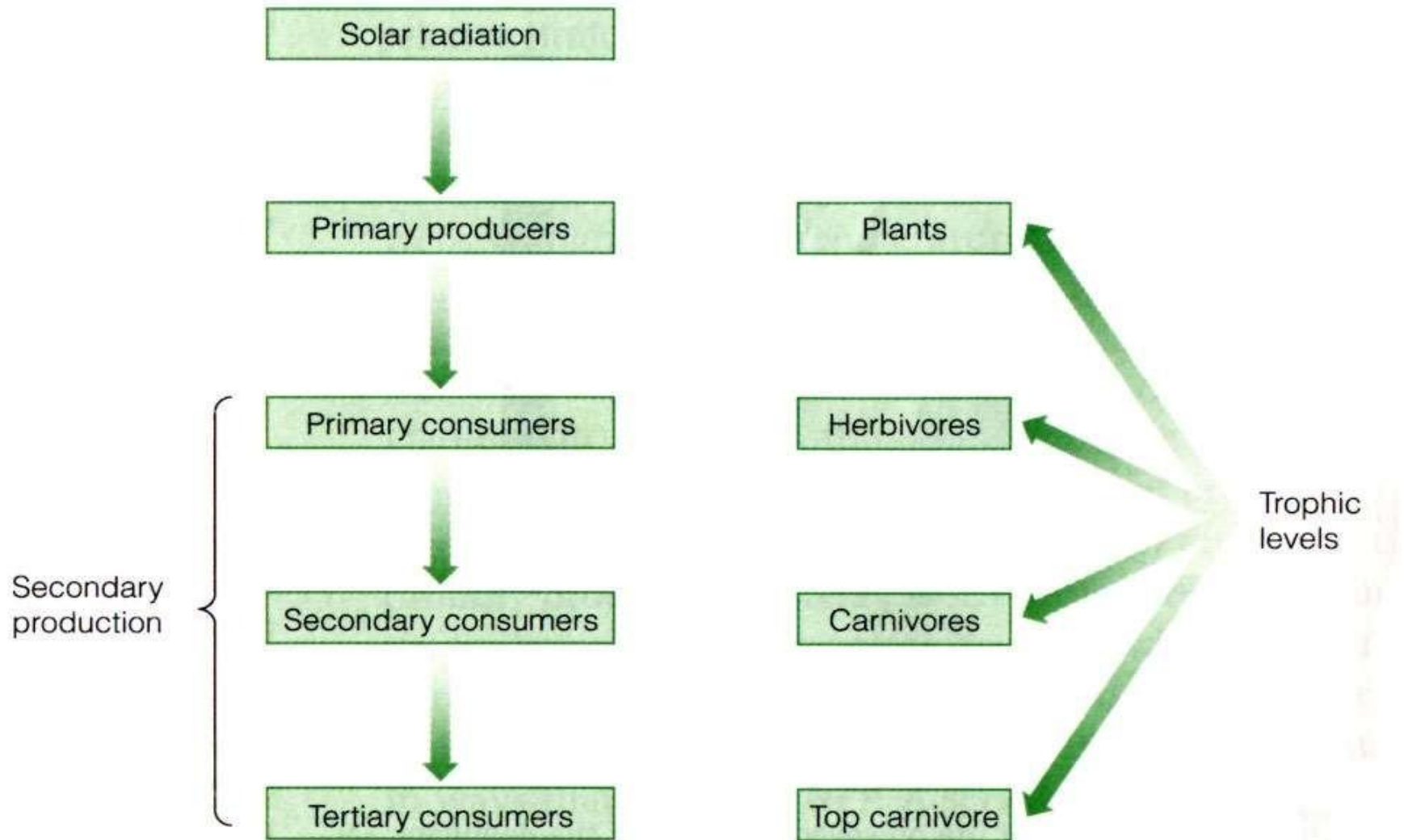
Tertiary Consumer
(Carnivore)

Secondary Consumers
(Carnivores or Omnivores)

Primary Consumers
(Herbivores)

Producers
(Autotrophs)





The schematic structure of a food chain. Each trophic level may contain many species.

ECOLOGICAL SUCCESSION

In a particular area, one community of species may be replaced by another community; the **progressive replacement of one community by another** till the development of stable community in a particular area is called as **ecological succession**.

TYPES OF SUCCESSION

1. Primary succession.

If an area in any of the basic environments (such as terrestrial, fresh-water or marine) is colonized by organisms for the first time, the succession is called **primary succession**.

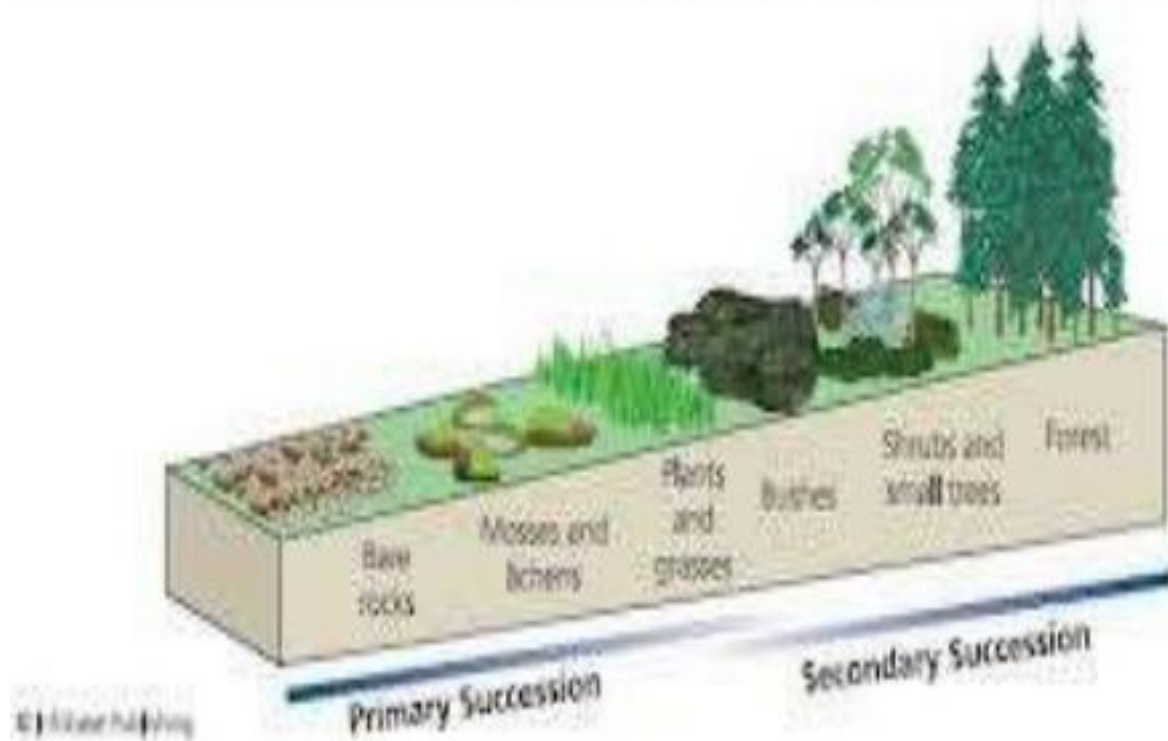
2. Secondary succession.

If the area under colonization has been cleared by whatsoever agency (such as burning, grazing, clearing, felling of trees, sudden change in climatic factors, etc.) of the previous plants, it is called **secondary succession**.

3. Autogenic succession.

After the succession has begun, in most of the cases, it is the community itself (as a result of its reactions with the environment) modifies its own environment and, thus, causing its own replacement by new communities. This course of succession is known as **autogenic succession**.

Ecological Succession



FOOD CHAIN IN ECOSYSTEM

- In an ecosystem one can observe the transfer or flow of energy from one trophic level to other in succession.
- **A trophic level** can be defined **as the number of links by which it is separated from the producer**, or as the which position of the organism in the food chain.
- Thus, primary producers trap radiant energy of sun and transfer that to chemical or potential energy of organic compounds such as carbohydrates, proteins and fats.
- When a herbivore animal eats a plant (or when bacteria decompose it) and these organic compounds are oxidized, the energy liberated is just equal to the amount of energy used in synthesizing the substances (first law of thermodynamics), but some of the energy is heat and not useful energy (second law of thermodynamics).

- If this animal is eaten by another one, along with transfer of energy from a herbivore to carnivore a further decrease in useful energy occurs as the second animal (carnivore) oxidizes the organic substances of the first (herbivore or omnivore) to liberate energy to synthesize its own cellular constituents.
- Such transfer of energy from organism to organism sustains the ecosystem and when energy is transferred from individual to individual in a particular community, as in a pond or a lake or a river, we come across the food chains.



Sun



Hawk

Mouse



Snake



Grass





Carnivore



Carnivore



Carnivore



Herbivore



Plant

**Quaternary
consumers**

**Tertiary
consumers**

**Secondary
consumers**

**Primary
consumers**

**Primary
producers**

A terrestrial food chain



Carnivore



Carnivore



Carnivore



Zooplankton



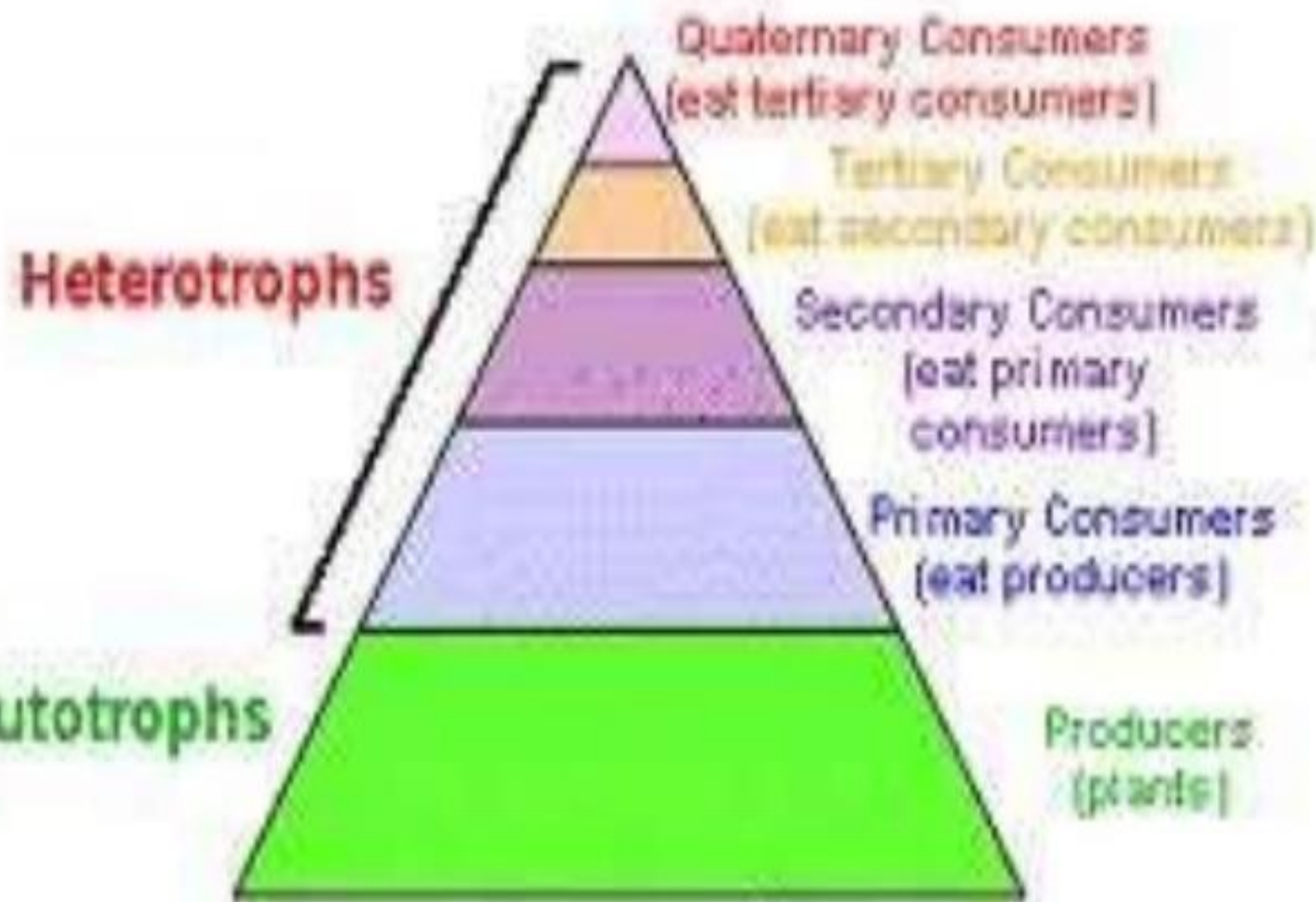
Phytoplankton

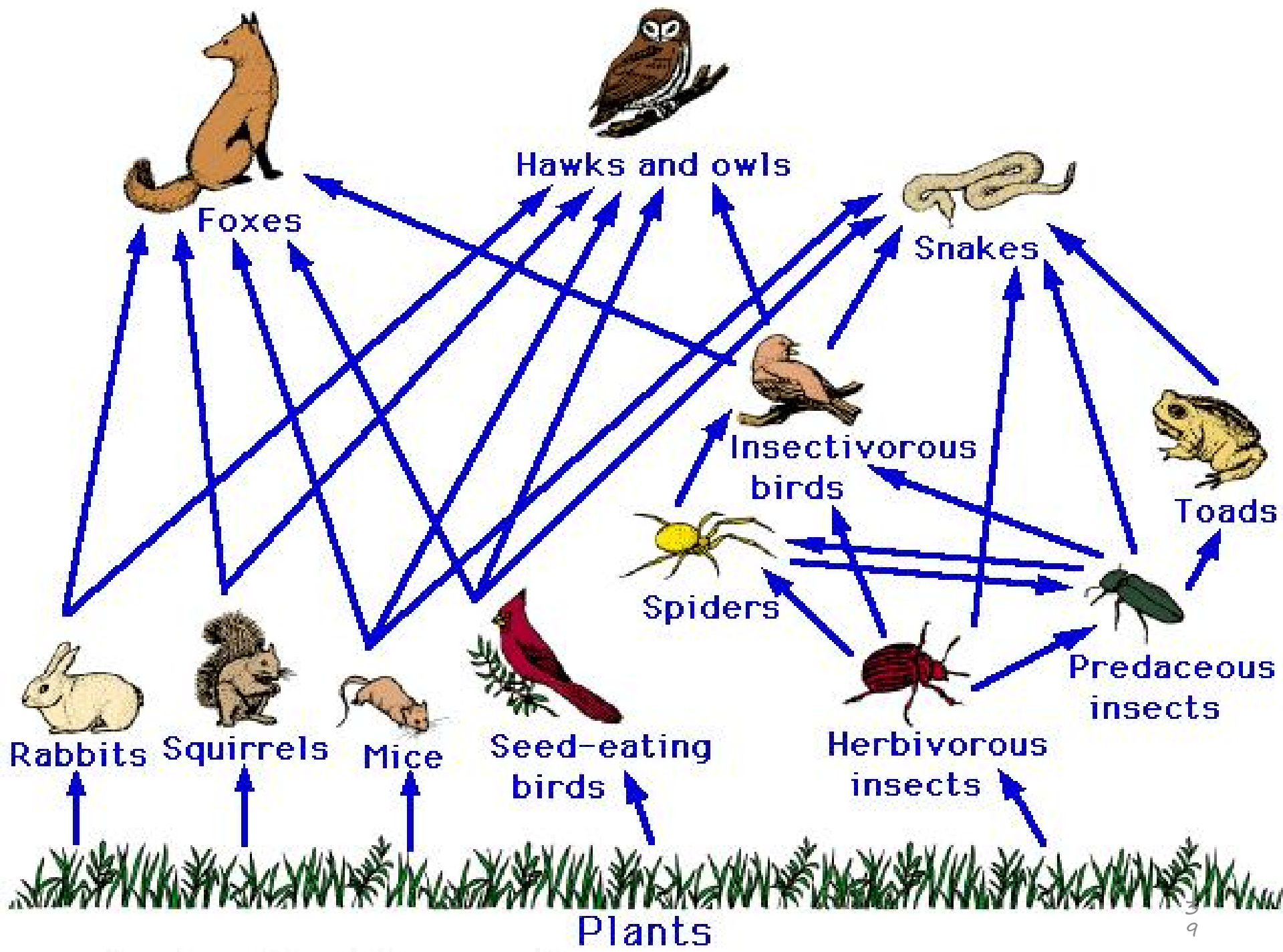
A marine food chain

FOOD WEB

In nature simple food chains occur rarely. The same organism may operate in the ecosystem at more than one trophic level i.e. it may derive its food from more than one source. Even the same organism may be eaten by several organisms of a higher trophic level or an organism may feed upon several different organisms of a lower trophic level. usually the kind of food changes with the age of the organism and the food availability. Thus in a given ecosystem various food chains are linked together and interrelated each other to form a complex network called food Web.

The Food Web





Significance of food chains and food webs

1. Energy flow and nutrient cycling takes place through them.
2. They maintain and regulate the population size of different trophic levels, and thus help in maintaining ecological balance.
- 3. Biomagnification:** The non-biodegradable materials keep on passing from one trophic level to another. At each successive trophic level, the concentration keeps on increasing. This process is known as biomagnification.

ex. DDT — (pesticide) The concentration is increased along the food chain through phytoplankton to zooplanktons and then goes to fish, birds, animals and human beings. DDT damages the shells in the bird's eggs and cells of the human beings

Ecological Pyramids

In the successive steps of grazing food chain-photosynthetic autotroph, herbivorous heterotroph, carnivores heterotroph, decay bacteria-the number and mass of the organisms in each step is limited by the amount of energy available. Since some energy is lost as heat, in each transformation the steps become progressively smaller near the top. This relationship is sometimes called **"ecological pyramid"**. The ecological pyramids represent the trophic structure and also trophic function of the ecosystem.

Types of Ecological Pyramids

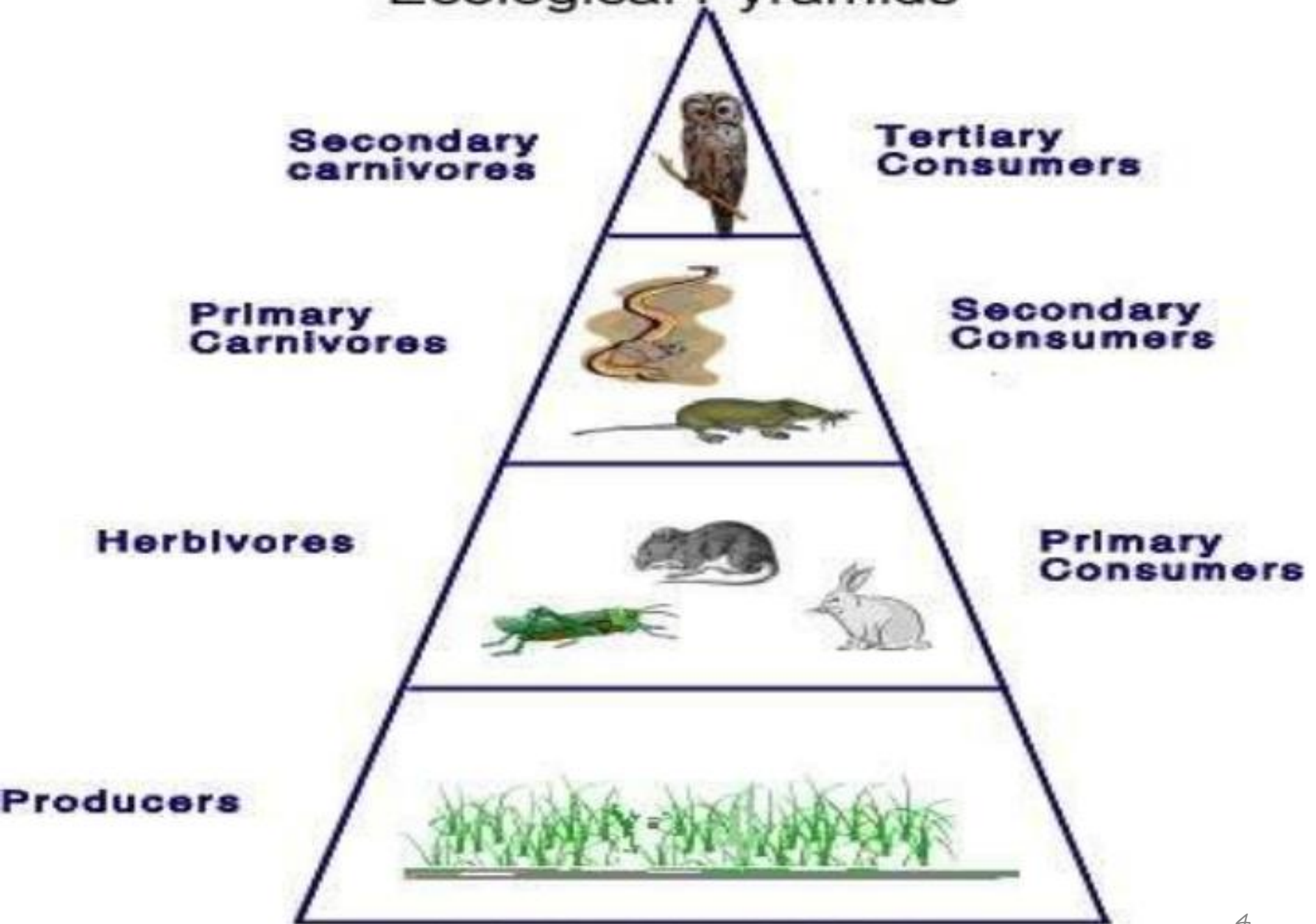
The ecological pyramids may be of following three kinds

- **Pyramid of number.**
- **Pyramid of biomass.**
- **Pyramid of energy.**

Pyramid of number

It depicts the number of individual organisms at different trophic levels of food chain. The animals at the lower end (base of pyramid) of the chain are the most abundant. Successive links of carnivores decrease rapidly in number until there are very few carnivores at the top. The pyramid of number ignores the biomass of organisms and it also does not indicate the energy transferred or the use of energy by the groups involved. The lake ecosystem provides a typical example for pyramid of number.

Ecological Pyramids



Pyramid of biomass

The biomass of the members of the food chain present at any one time forms the pyramid of the biomass. Pyramid of biomass indicates decrease of biomass in each trophic level from base to apex. For example, the total biomass of the producers ingested by herbivores is more than the total biomass of the herbivores in an ecosystem. Likewise, the total biomass of the primary carnivores (or secondary consumer) will be less than the herbivores and so on.

BIOMASS
(g/m²)

TROPHIC LEVEL

10



Tertiary consumers
(snakes)

100



Secondary consumers
(toads)

1,000



Primary consumers
(grasshoppers)

10,000



Producers
(grass)

Pyramid of energy

When production is considered in terms of energy, the pyramid indicates not only the amount of energy flow at each level the actual role the various organisms play in the transfer of energy. the pyramid of energy is constructed is the quantity of organisms produced per unit time.

Tertiary
consumers



10 J

Secondary
consumers



100 J

Primary
consumers



1,000 J

Primary
producers



10,000 J

1,000,000 J of sunlight

Classification of Ecosystems

- **Terrestrial Ecosystems** -which encompass the activities that take place on land
- **Aquatic ecosystems** - the system that exists in water bodies

Terrestrial ecosystem

- ❖ Forest ecosystem
- ❖ Desert ecosystem
- ❖ Grassland ecosystem

Aquatic ecosystem

- ❖ Marine ecosystem
- ❖ Fresh water ecosystem
- ❖ Estuarine ecosystem

FOREST ECOSYSTEM

Undisturbed areas with moderate to high average annual rain precipitation tend to be covered with forest, which contains various species of trees and smaller forms of vegetation.

There are three important types of forests are

- ❖ Tropical rain forests.
- ❖ Temperate deciduous forests.
- ❖ Coniferous forests.

Structure (components) and function of forest ecosystem

I. Abiotic components:

The abiotic components are physical (inorganic and organic substances) components found in soil and atmosphere. (E.g. Temperature, light, rain fall)

II. Biotic components:

1. **Producers:** The plants absorb sunlight and produce food through photosynthesis.(e.g. Trees, shrubs and ground vegetation)
2. **Consumers:**
 - a) *Primary consumers*: They directly depend on the plants for their food.(e.g.. Ants, flies, insects, mice, deer)
 - b) *Secondary consumers*: They directly depend on the herbivores for their food. (e.g. Snakes, birds, fox)
 - c) *Tertiary consumers*: They depend on the primary carnivores for their food.(e.g. .Tiger, lion)
3. **Decomposers:** Rate of decomposition in tropical and subtropical forests is more rapid than in the temperate forests. (e.g. Bacteria and fungi).

GRASSLAND ECOSYSTEM

- Grasslands are regions with enough average annual rain precipitation to allow grass to grow extensively. But drought and fire does not allow trees to grow taller.
- Grasslands are rich biological communities of grasses, seasonal flowering plants and open savannas.
- Great Plains of central North America, Russia, and South American are some of the important grasslands in the world

Structure and function of the grassland Ecosystems:

I. Abiotic components: These abiotic components are supplied by CO_2 (Nutrients C,H,O,N,P,S,etc.)

II. Biotic components:

1. Producers: They produce food.(Grasses, forbs and shrubs)

2. Consumers:

- a) **primary consumers:** They depend on grasses for their food.(Cows, buffaloes, deer, sheep, etc.,)
- b) **secondary consumers:** They feed on herbivores.(Snakes, lizards, birds, jackals, fox, etc.,)
- c) **Tertiary consumers:** They feed on secondary consumers.(Hawks, eagles, etc.,)

3. Decomposers: Fungi and bacteria (decompose the dead organic matter.

DESERT ECOSYSTEM

- Deserts are dry places with unpredictable and infrequent precipitation. A desert is an area where evaporation exceeds precipitation.
- Daily and seasonal temperature of the desert will vary since desert has very little moisture to absorb and store sun radiation.
- Deserts with less than 2.5 cm of precipitation supports almost zero vegetation.
- Deserts with 2.5 to 5.0 cm precipitation have thin (scanty) vegetation (less than 10% of the ground is covered).
- Seasonal leaf production, water-storage tissues and thick epidermal layer help to reduce water loss.
- A combination of low rainfall and different average temperatures creates tropical, temperate and cold deserts.

Structure and functions of the desert ecosystems:

I. **Abiotic components:** Temperature water, sunlight, rainfall, etc. The temperature is very high and low rain fall & low nutrient cycling.

II **Biotic components:**

1. **Producers:** Shrubs, bushes

In deserts mostly Succulent (e.g., cacti) plants are found available. They have water inside them to stay alive.

2. **Consumers:** Squirrels, mice, rabbits, reptiles.

These animals dig holes in the ground to live in. Most of the animals can extract water from the seeds they eat.

3. **Decomposers:** Fungi and bacteria

Desert has poor vegetation with a very low amount of dead organic matter.

AQUATIC (Water) ECOSYSTEMS

Freshwater Ecosystems

Wetlands:

Once considered useless, disease ridden places (e.g., malaria and yellow fever)

Provide many benefits to society:

- ❖ Fish and wildlife habitats
- ❖ Natural water quality improvement
- ❖ Flood storage

They also are a source of substantial biodiversity in supporting major groups of organisms numerous species from all of the from microbes to mammals

Aquatic Ecosystems



Structure and function of aquatic ecosystem

I. **Abiotic components**: Temperature, light proteins, O₂

II. **Biotic components**:

1. Producers: They are green plants, may be submerged, free floating and amphibious plants. phytoplanktons, algae and flagellates

2. Consumers: a) **Primary Consumers** (Zooplanktons)
Cilicetes, protozoans, etc., - They feed on phytoplankton.

b) **Secondary consumers** (carnivores) Insects and small fishes. – They feed on zooplankton.

c) **Tertiary consumers**: Large fishes like game fish - They feed on smaller fish.

1. Decomposers: Bacteria, fungi and actinomycetes.- They decompose the dead plants and animals.

Biodiversity and its conservation

- **Biodiversity is the variety of life on earth It includes all life forms-from the unicellular fungi, protozoa and bacteria to complex multi-cellular organisms such as plants, birds, fishes and animals.**
- **Biodiversity is the variety of flora and fauna on this planet earth.**
- **The concept reflects the inter-relatedness of genes, species and ecosystems. Because genes are the components of species, and species are the components of ecosystems**
- **Diversity may be defined as the number of species present in a community, a measure termed as species richness**



GENETIC DIVERSITY

- It refers to the total genetic information contained in the genes of individuals of plants, animals and microorganisms.
- The genes found in organisms can form enormous number of combinations each of which gives rise to some variability. Genes are the basic units of hereditary information transmitted from one generation to other.
- When the genes within the same species show different versions due to new combinations, it is called genetic variability.



SPECIES DIVERSITY

- A species generally consists of all the individual organisms of a natural population which are able to interbreed, generally sharing similar appearance, characteristics and genetics. A species is one of the basic units of biodiversity.
- Species richness is the simplest measure of biodiversity and is simply a count of the number of different species in a given area.

- Species evenness is a diversity index, a measure of biodiversity which quantifies how equal the populations are numerically
- This is the variability found within the population of a species or between different species of community.
- It represents broadly the species richness and their abundance in a community.



ECOSYSTEM DIVERSITY

This is the diversity of ecological variations in

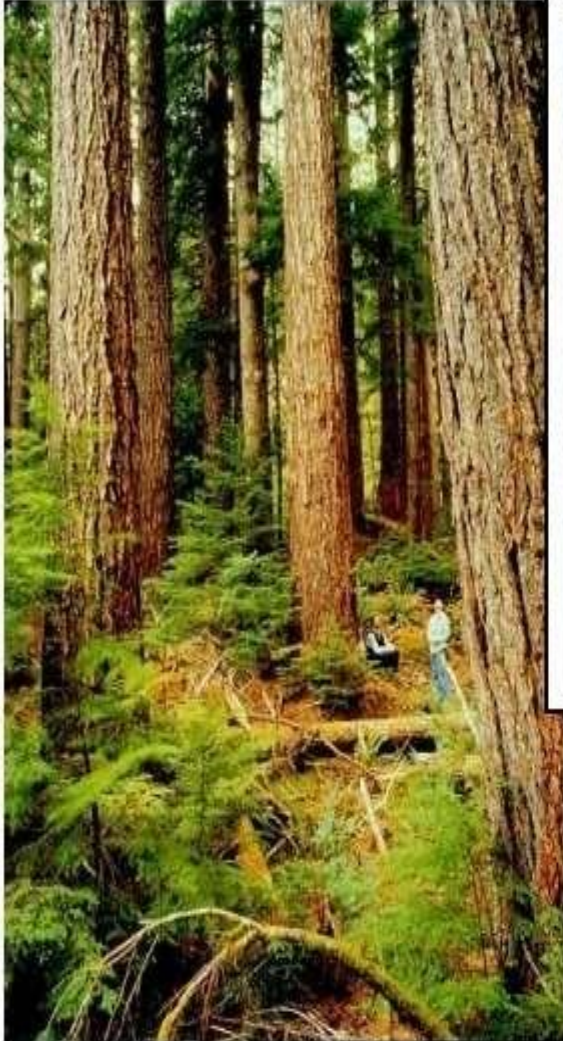
- tropic structure,
- food-webs,
- nutrient cycling etc.

The ecosystems also show variations with respect to physical parameters like

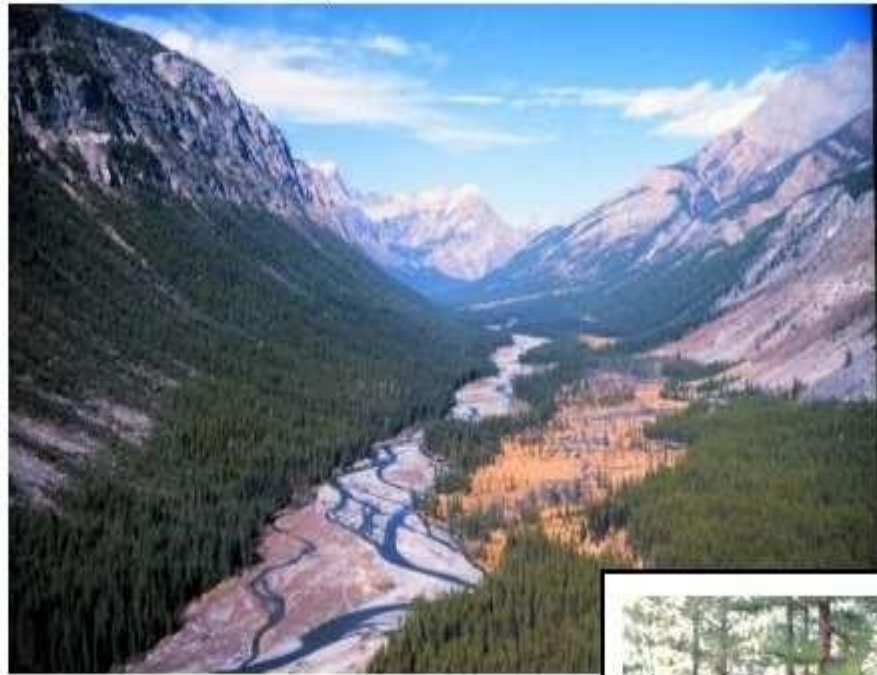
- moisture
- temperature
- altitude
- precipitation etc.,

- The ecosystem diversity is of great value that must be kept intact. This diversity has developed over millions of years of evolution. If we destroy this diversity, it would disrupt the ecological balance.
- We cannot even replace the diversity of one ecosystem by that of another. Coniferous trees of boreal forests cannot take up the function of the trees' of tropical deciduous forest lands and vice versa

Palliser River Valley



Coastal forest



Dry open forest

Value of Bio-Diversity

Consumptive value:

These are direct use values where the biodiversity product can be harvested and consumed directly e.g. fuel, food, drugs, fibre etc.

Drugs and medicines:

- About 75% of the world's population Depends upon plants or plant extracts for medicines. The wonder drug Penicillin used as an antibiotic is derived from a fungus called "Penicillium.
- we get Tetracyclin from a bacterium. Quinine, the cure for malaria is obtained from the bark of Cinchona tree,
- Digitalin is obtained from foxglove (Digitalis) which is an effective cure for heart disease.
- Vinblastin and vincristine, two anticancer drugs, have been obtained from Periwinkle (Catharanthus) plant, which possesses anticancer alkaloids.

Fuel:

Our forests have been used since ages for fuel wood. The fossil fuels coal, petroleum and natural gas are also products of fossilized biodiversity.

PRODUCTIVE VALUES:

These are the commercially usable values where the product is marketed and sold. These may include the animal products like tusks of elephants, musk from musk deer, silk from silkworm, wool from sheep, fur of many animals etc, all of which are traded in the market. A wild variety of rice from UP. Saved millions of hectares of paddy crop from Grosby-Stuntvirus

SOCIAL VALUES/ETHICAL VALUES:

- ❖ These are the values associated with the social life, customs, religion and aspects of the people.
- ❖ Many of the plants are considered holy and sacred in our country like Tulsi (holy basil),Peepal, Mango, Lotus, Neem etc.
- ❖ The leaves, fruits or flowers of these plants are used in worship or the plant itself is worshipped. The tribal people are very closely linked with the wild life in the forests.

AESTHETIC VALUE:

- Great aesthetic value is attached to biodiversity. We will not like to visit vast barren lands with no signs of visible life.
- People from far and wide spend a lot of time and money to visit wilderness areas where they can enjoy the aesthetic value of biodiversity and this type of tourism is now known as **ecotourism**.

ECOLOGICAL VALUE:

It refers to the services provided by ecosystems such as

- ❖ prevention of soil erosion,
- ❖ prevention of floods,
- ❖ maintenance of soil fertility,
- ❖ nutrients cycles,
- ❖ fixation of nitrogen,
- ❖ hydrological cycle,
- ❖ acts as carbon sinks,
- ❖ pollutant absorption
- ❖ Reduction of the threat of global warming

Threats to Bio-Diversity

Extinction, the elimination of a species is a normal process in nature. Species however, human impacts on populations and ecosystems have accelerated that rate of extinction, causing hundreds of species, sub-species and varieties to become extinct every year.

The causes of extinction are:

Population Risk: uneven change in population rates (i.e. birth rates and death rates) can cause a species in low abundance to become extinct. e.g. blue whales.

Environmental Risk: Environmental risk means variation in the physical or biological environment, including variations in predator, prey, symbiotic or competitor species. In case of species that are sufficiently rare and isolated, such normal environmental variations can lead to their extinction.

Natural calamities: A natural catastrophe is a sudden change in the environment .It includes fires, storms, floods, earthquakes, volcanic eruptions etc. Such a natural phenomenon may cause the extinction of most forms of life there.

Genetic Risk: Change in genetic characteristics in a small population of a species, due to reduced genetic variation, genetic drift or mutation, genetic assimilation [cross-breeding] makes the species more vulnerable to extinction.

Human Actions: human activities like hunting , development of agriculture, rise of civilization, rapid deforestation and introduction of industrial chemicals and emissions, pesticides and pollution are also leads to extinction of species.

Habitat Loss and Degradation: Habitat loss and degradation are the major proximate causes of species extinction, affecting 89% of all threatened birds, 83% of mammals and 91% of all threatened plants assessed globally. The main causes of habitat loss are agricultural activities, harvesting or extraction (including mining, fishing, logging, etc.) and development of human settlements, industry and associated infrastructure.

Diseases: Pathogens, or disease organisms, may cause extinction. Animals [in sanctuaries and reserves] are more prone to infection when they are under stress.

Poaching: Poaching is another threat

Endangered Species of India

- ❖ An **endangered species** is a population of organisms which is at risk of becoming extinct because it is either few in numbers, or threatened by changing environmental or predation parameters.
- ❖ In India, nearly 450 plant species have been identified in the categories of endangered, threatened or rare. Existence of about threatened while an unknown number of species of insects relevance here to give a complete list of endangered flora and fauna of our country. However, a few species of endangered reptiles, birds, mammals and plants are given below:

Reptiles:

- ❖ green sea turtle,
- ❖ tortoise,
- ❖ python

Birds:

- ❖ Great Indian bustard,
- ❖ Peacock, Pelican,
- ❖ Siberian White Crane

Carnivorous:

- ❖ Indian wolf,
- ❖ Mammals tiger,
- ❖ leopard,
- ❖ striped hyena,
- ❖ Indian lion

Conservation of Bio-Diversity

- ❖ The act or process of conserving. The protection, preservation, management, or restoration of wildlife and of natural resources such as forests, soil, and water.

Conservation of our natural resources has the following three specific objectives:

- ❖ (i) to maintain essential ecological processes and life-supporting systems .
- ❖ (ii) to preserve the diversity of species or the range of genetic material found in the organisms on the planet .
- ❖ (iii) to ensure sustainable utilization of species and ecosystems which support millions of rural communities as well as the major industries all over the world.

There are two approaches of biodiversity conservation:

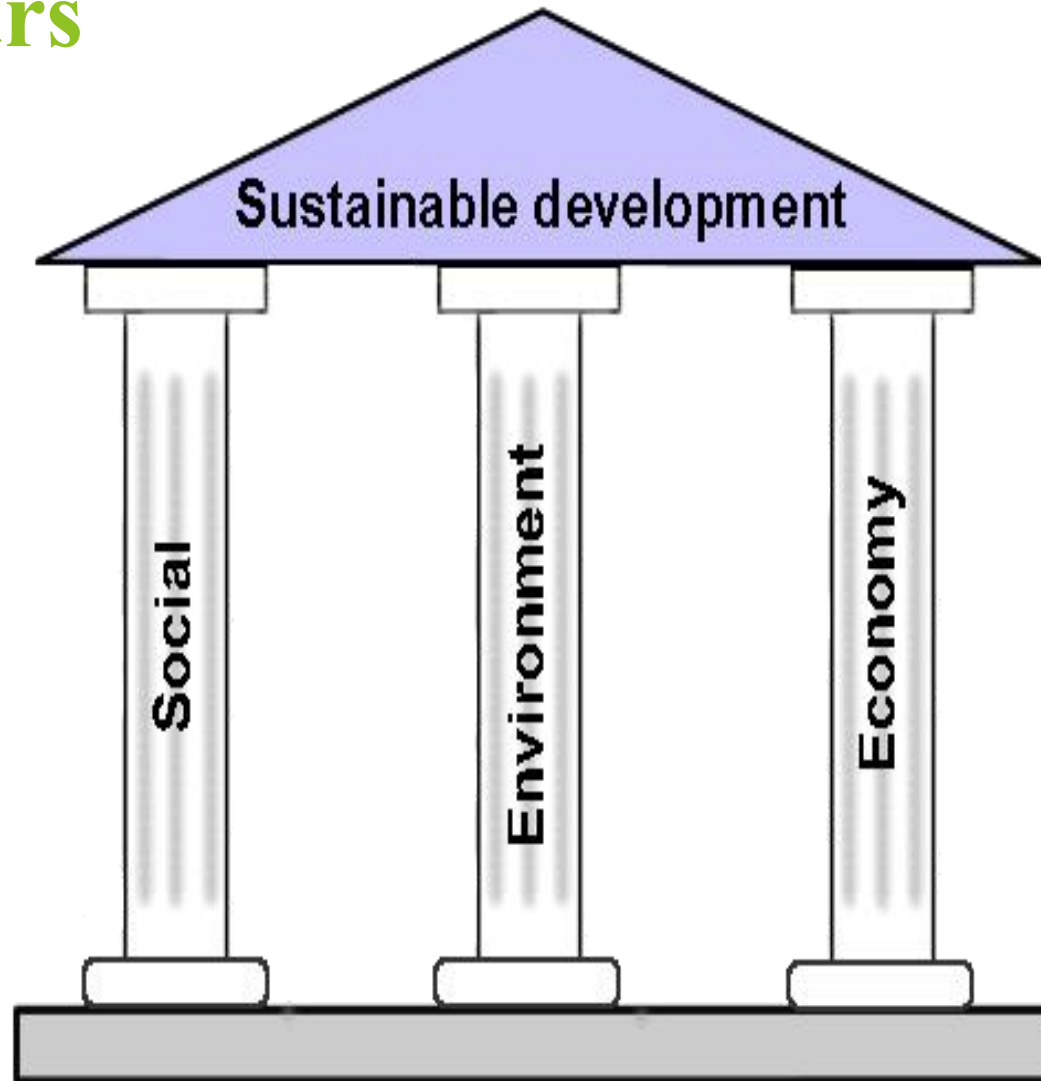
- ❖(a) In situ conservation (within habitat): This is achieved by protection of wild flora and fauna in nature itself, e.g. Biosphere Reserves, National Parks, Sanctuaries, Reserve Forests etc.
- ❖(b) Ex situ conservation (outside habitats) This is done by establishment of gene banks, seed banks, zoo, botanical gardens, culture collections etc.

Definition of Sustainable Development

According to the World Commission on Environment and Development, the definition of Sustainable Development (SD) is “development that meets the needs of the present without compromising the ability of future generations to meet their own need” (Our Common Future, 1987).

3 Dimensions

Sustainability - 3 pillars



What makes SD

- **Social Goals**

Sustains institutions , improves justice and encourages participation

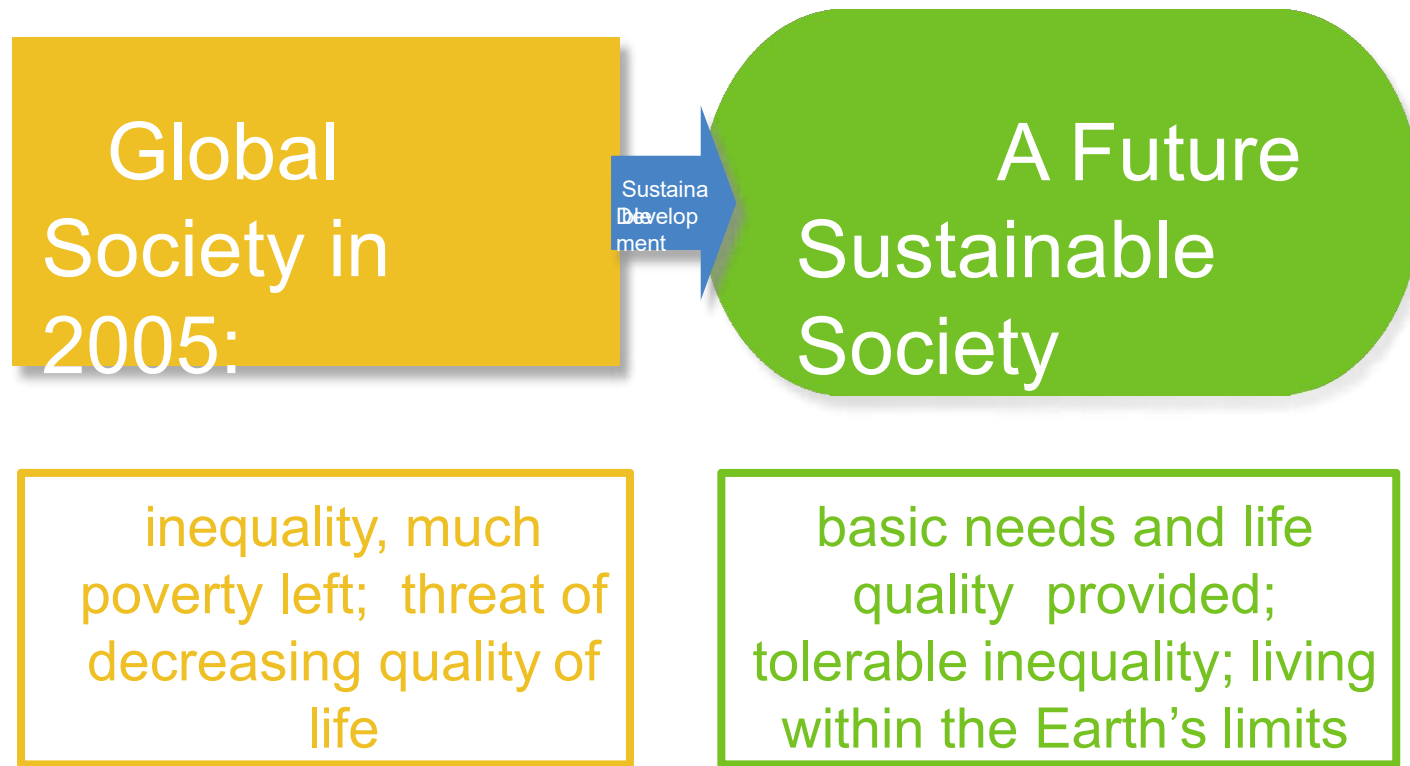
- **Economic Goals**

Ensures basic needs, equity and employment opportunities

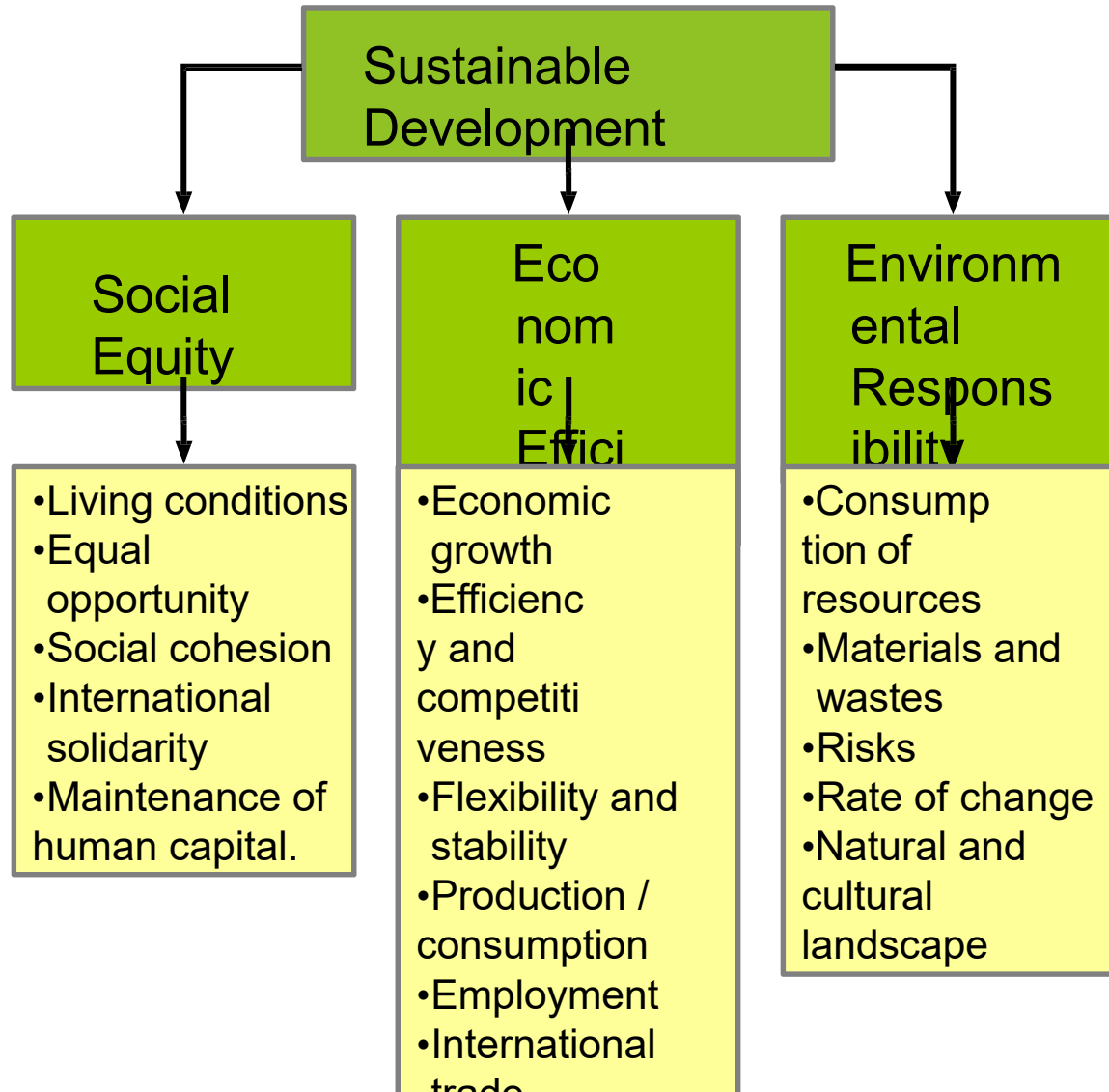
- **Environmental Goals**

Maintains genetic diversity which maximizes productivity and renewal

Sustainable Development is the journey *towards* Sustainability

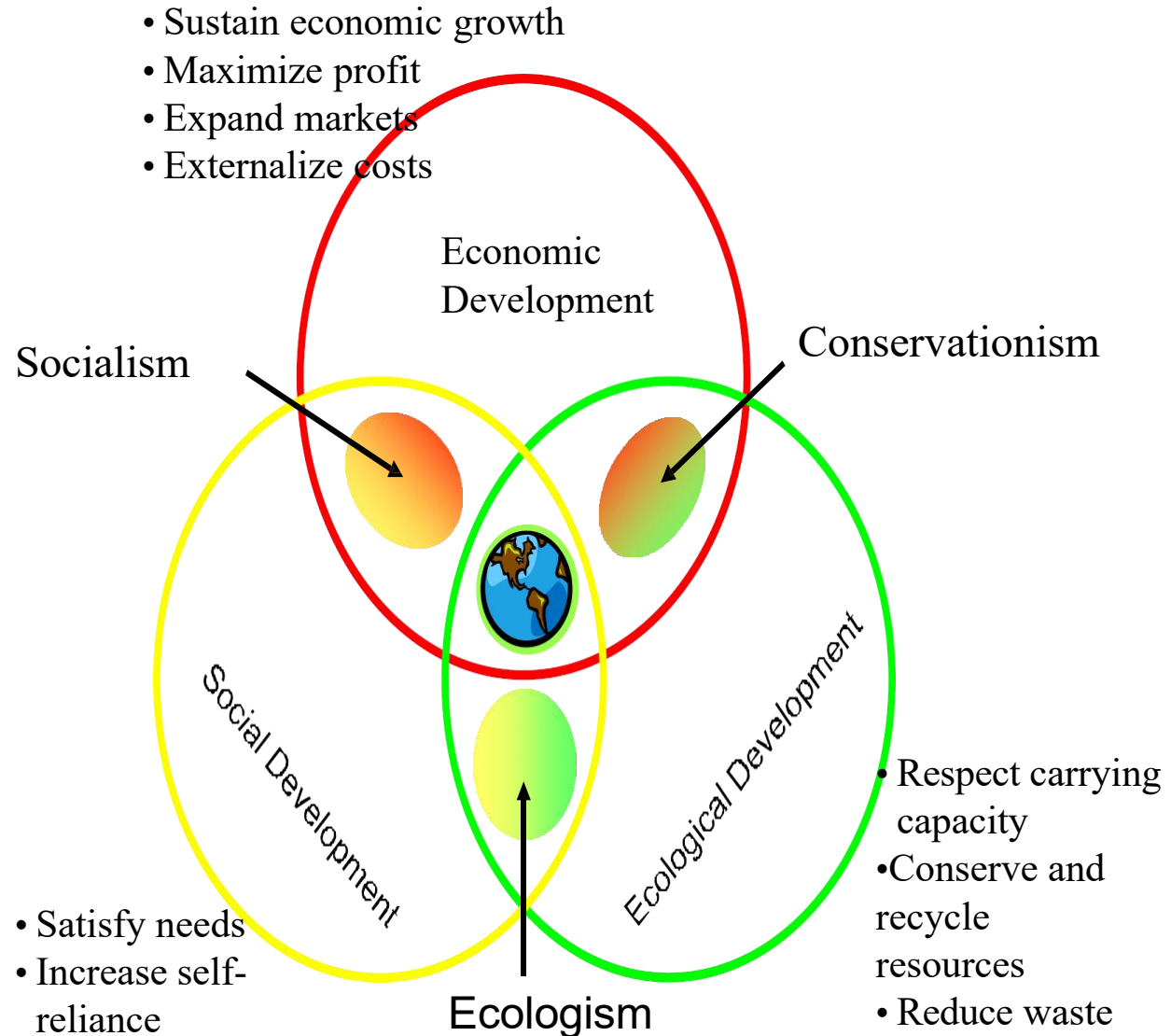


Sustainable Society



3 Approaches

Global Sustainability



The three core drivers of unsustainability

Consumption

Use of resources beyond the reasonable limits set by nature

Production

Gross inefficiencies in production.

Distribution

Inequitable distribution e.g. distribution of global income between rich and poor

Solution

“The significant problems we face cannot be solved at the same level of thinking we were at when we created them.” - Albert Einstein

Sustainability: Solutions

- ◆ Cyclical material use
 - emulate natural cycles; 3 R's
- ◆ Safe reliable energy
 - conservation, renewable energy, substitution, interim measures
- ◆ Life-based interests
 - health, creativity, communication, coordination, appreciation, learning, intellectual and spiritual development

Conclusion

Is a long journey towards SD

End-point not definable

Each country/region has different goals, based on economic circumstances, population size, stage of development, resource endowment etc....